Causation and Research Design

POL 51: Scientific Study of Politics

Spring 2026

Causality

- We want to evaluate causal relationships in social science: $X \to Y$.
- Hypotheses are phrased as causal statements:
  - Get-out-the-vote mailers $\to$ voter turnout.
  - Age $\to$ political ideology.
  - Democracy $\to$ GDP.

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  - Get-out-the-vote mailers $\to$ voter turnout.
  - Age $\to$ political ideology.
  - Democracy $\to$ GDP.
- Is there a statistically significant **causal** relationship between the two concepts, and is it meaningful?
- The first variable (the **cause**) is referred to as: X, the independent variable(s), or the right hand side (RHS) variable.
- The second variable (the **effect**) is referred to as: Y, the dependent variable, or the left hand side (LHS) variable.

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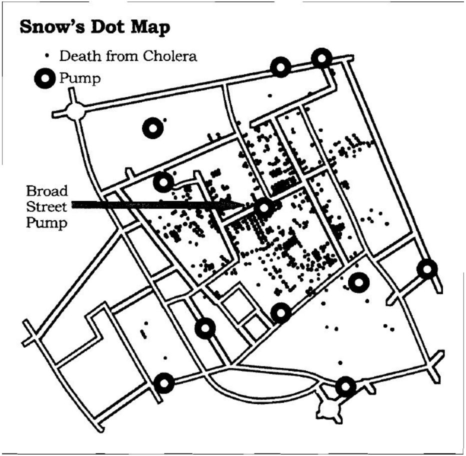
  - For example, shark attacks and ice cream sales.
  - What about something like campaign spending and vote % (*Freakonomics*)?

Causal inference: Cholera in 19th Century London

Miasmitic theory: bad air → cholera.

Causal inference: Cholera in 19th Century London

John Snow: found concentration of deaths around Broad St. pump. Turns out, water company drawing water from polluted Thames. Concluded bad water $\rightarrow$ cholera.



Causal inference: Cholera in 19th Century London

- Evidence like this led to transition from miasmitic theory to germ theory.
- Quick aside: the one major anomaly in Snow's finding was that none of the brewery workers near Broad St contracted cholera.

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  4. Control for confounding variables
     - What variables might be associated with both life satisfaction and democratic stability?

Research Design

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- Experimental vs. observational studies.

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- Example: how does music influence shopping behavior at supermarkets?
- Quasi-experiments: field experiments and natural experiments.
- Problems: low external validity and difficult to construct.

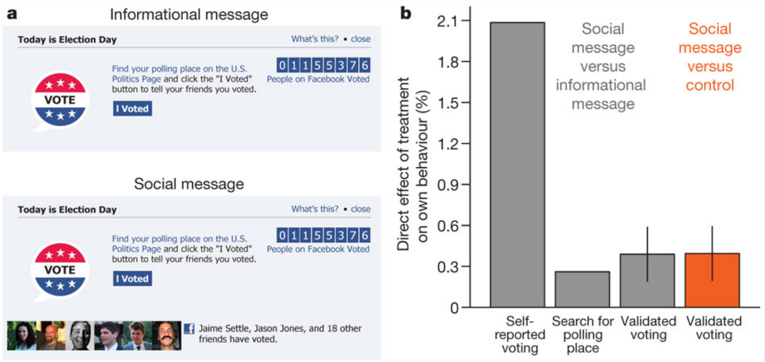
Figure 1: **The experiment and direct effects.**

From

A 61-million-person experiment in social influence and political mobilization

Robert M. Bond, Christopher J. Fariss, Jason J. Jones, Adam D. I. Kramer, Cameron Marlow, Jaime E. Settle & James H. Fowler

Nature **489**, 295–298 (13 September 2012) | doi:10.1038/nature11421



a, **b**, Examples of the informational message and social message Facebook treatments (**a**) and their direct effect on voting behaviour (**b**). Vertical lines indicate s.e.m. (they are too small to be seen for the first two bars).

Observational Studies

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- Use real-world data to evaluate causal relationships in political science.
- Cross-sectional data and time series data.
- Problems: low internal validity because of difficulties in controlling for confounders (no random assignment).

Observational data

Data Editor (Edit) - [anes_pilot_2016]

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info[9] 4

	version	caseid	weight	weight_spss	follow	turnout12	turnout12b	vote12	percent16	meet	givefut	info	
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2	ANES 2016 Pilot Study version 20160223	2	2.6701962	1.5220118	Some ...	Defin...	.	.	50	A lit...	Not a...	A lit...	H
3	ANES 2016 Pilot Study version 20160223	3	1.4303896	.81532206	Most ...	Defin...	.	Mitt ...	100	Extre...	Extre...	Extre...	H
4	ANES 2016 Pilot Study version 20160223	4	.91396618	.52096072	Most ...	Defin...	.	Barac...	100	Not a...	A lit...	Not a...	H
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8	ANES 2016 Pilot Study version 20160223	8	.31802712	.18127546	Most ...	Defin...	.	Mitt ...	100	Not a...	A lit...	Not a...	H
9	ANES 2016 Pilot Study version 20160223	9	1.0194665	.58109593	Most ...	Defin...	.	Mitt ...	100	Moder...	Moder...	A lit...	H
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17	ANES 2016 Pilot Study version 20160223	17	.62891779	.35848314	Most ...	Defin...	.	Barac...	100	Extre...	Extre...	Extre...	H

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